

Agriculture Profitability Intelligence Dashboard

Interactive • Dark
Agriculture-Tech
Theme

PROJECT STACK

Power BI
Power Query
DAX
Excel

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THE BUSINESS PROBLEM

- Farm profitability changes with rainfall, fertilizer usage, production cost and market price.
- Comparing crop, state and season performance manually is slow and inconsistent.
- Decision-makers need one view of profitability drivers and performance.
- The objective is to identify the most profitable Crop, State, Season and Irrigation Type.
- Power BI converts agricultural records into an interactive decision-support dashboard.

CORE QUESTION

Where should
farmers focus
to maximize
profit?

DATA OVERVIEW

RECORDS

500

COLUMNS

13

YEARS

2021-2024

TARGET

Profit /
Acre

DATA DOMAINS

LOCATION

State, District

AGRONOMY

Crop, Season, Yield, Rainfall,
Soil

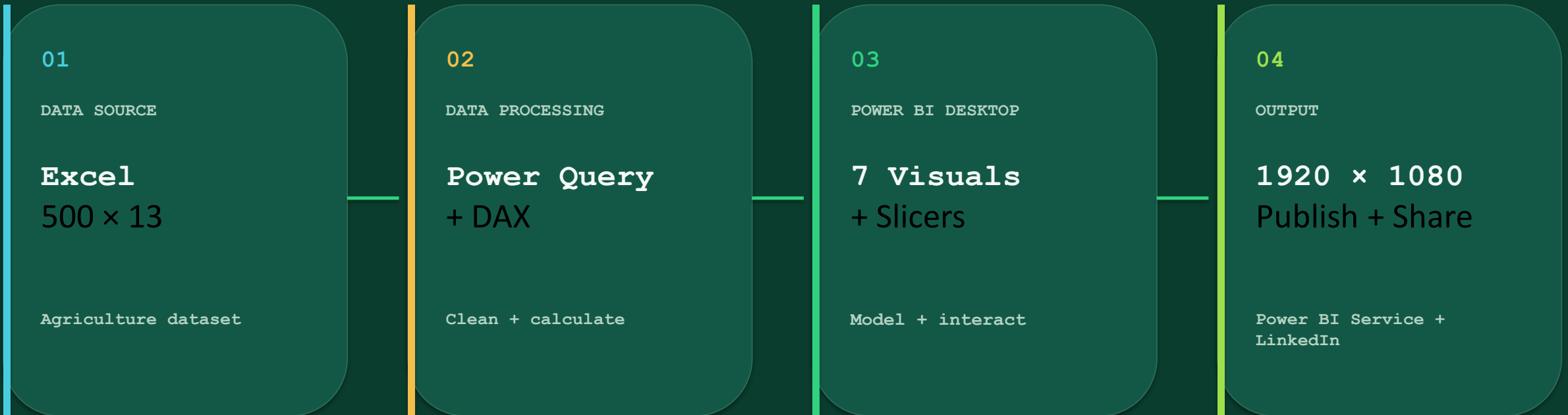
INPUTS

Fertilizer Usage, Irrigation Type

ECONOMICS

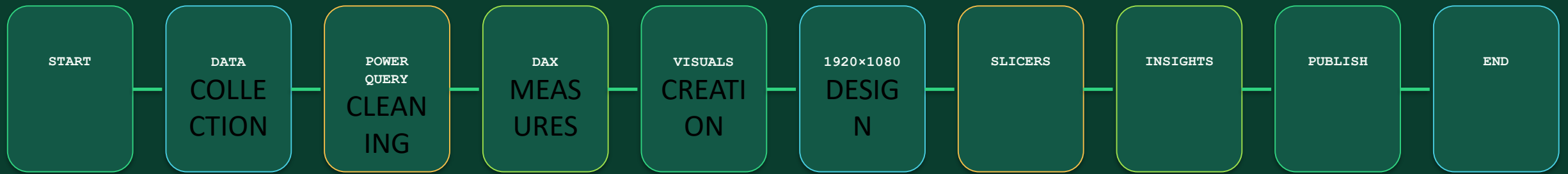
Market Price, Production Cost,
Profit

SOLUTION ARCHITECTURE



DAX examples: $\text{Total Profit} = \text{SUM}(\text{Profit}) \cdot \text{Avg Yield} \cdot \text{Profit Margin \%} = \text{Profit} / \text{Cost} \times 100$

ANALYTICS FLOW



DATA → QUALITY → MEASURES → VISUAL STORY → DECISION

DASHBOARD OVERVIEW

POWER BI DASHBOARD SCREENSHOT

TOTAL PROFIT

AVG YIELD

AVG PROFIT

TOTAL COST

STATE-WISE PROFIT MAP

CROP-WISE PROFIT BAR

YEAR TREND + SEASON

KEY INSIGHTS

TOP CROP

N/A

TOP STATE

N/A

BEST SEASON

N/A

TOP IRRIGATION

N/A

TOP CROP AVERAGE PROFIT / ACRE

₹0

N/A ranks first by mean profit per acre.

CORRELATION SIGNALS

Rainfall ↔ Yield: 0.00

Fertilizer ↔ Profit: 0.00

Market Price ↔ Profit: 0.00

IMPACT

FARM DECISIONS

Choose stronger crop-state combinations using profitability evidence.

RESOURCE PLANNING

Compare irrigation, fertilizer and soil patterns to identify efficient practices.

AGRITECH VALUE

Support advisory services, product targeting and farmer segmentation.

POLICY PLANNING

Provide a scalable analytical layer for regional agricultural planning.

Scalable next: 10,000+ rows • real-time market prices • weather integration

RECOMMENDATIONS

01 IRRIGATION Evaluate the highest-performing irrigation option by crop and soil before adoption.

02 CROP + STATE Prioritize crop-state combinations with consistently high average profit per acre.

03 SEASON PLANNING Use the Rabi/Kharif/Zaid profitability ranking to guide seasonal planning.

04 INPUT OPTIMIZATION More fertilizer is not automatically better—balance input cost against yield and profit.

05 SOIL INTELLIGENCE Use Soil Type × Crop profitability to improve crop-selection recommendations.

CONCLUSION & NEXT STEPS

WHAT WAS DELIVRED

- Interactive profitability dashboard answering crop, state, season and irrigation questions.
- Power Query + DAX workflow for repeatable data preparation and business measures.
- Visual storytelling through map, bar, line, donut, scatter and KPI cards.
- Decision-oriented insights from 500 agricultural records.

FUTURE VISION

From reporting
to predictive
farm intelligence

THANK YOU